



Global Scenario of Antimicrobial Resistance: A Review of the Homoeopathic Perspective in Mitigating this Menace

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ABSTRACT

Antimicrobial resistance (AMR) has become a major global health challenge, primarily driven by unchecked and irrational usage of antibiotics in human healthcare, veterinary practice, agriculture, and environmental settings. The growing ineffectiveness of conventional antimicrobials against resistant pathogens necessitates exploring sustainable alternatives. Homoeopathy, a well-established complementary and alternative system of medicine, offers potential as a supportive or first-line approach, particularly in managing community-acquired infections. This review evaluates the capacity of homoeopathic interventions to minimize antibiotic reliance and thereby contribute to AMR mitigation. A systematic literature search across PubMed, Google Scholar, BASE, and Science.gov identified both pre-clinical and clinical studies assessing homoeopathy in infections commonly managed with antibiotics, such as respiratory, urinary, and gastrointestinal conditions. Evidence indicates that homoeopathy can stimulate immune mechanisms, lower infection recurrence, and in several studies, demonstrate outcomes comparable to conventional treatments. Beyond human health, its application in veterinary and agricultural sectors also suggest potential to curb antibiotic overuse and reduce environmental contamination. The historical and current role of homoeopathy in epidemic control further underscores its relevance for public health integration. In alignment with the WHO's "One Health" framework, evidence-based incorporation of homoeopathy could provide a complementary strategy in addressing the substantial morbidity and mortality associated with AMR, particularly across India and the Asia-Pacific. Strengthening this approach requires rigorous, large-scale clinical research to validate efficacy and guide policy adoption.

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Introduction

Multimorbidity, together with polypharmacy, adverse drug reactions, and antimicrobial resistance (AMR), has emerged as one of the most significant public health concerns of the modern era. Antimicrobials were once hailed as a breakthrough in advancing strategies against infectious pathogens, offering substantial relief from previously devastating illnesses. However, their widespread and often inappropriate use in human medicine, veterinary practice, agriculture, and the environment has resulted in excessive reliance and accelerated resistance. Over the last fifty years, antibiotics have been extensively deployed for treatment, prophylaxis, and growth promotion in animals. This overuse has created intense selective pressure, enabling microorganisms to rapidly adapt, often developing resistance within just a few years of a new drug's introduction¹.

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AMR occurs through genetic changes such as conjugation, transformation, or transduction, and its emergence has been strongly linked to human activities, particularly misuse of antimicrobial agents. It is now ranked among the top ten global health threats and categorized under ICD-10 code Z16². The consequences of irrational prescribing and consumption are profound, reducing therapeutic effectiveness, increasing treatment failures, prolonging illness, and driving higher healthcare costs. Diseases that were once easily controlled—such as pneumonia, tuberculosis, gonorrhoea, and salmonellosis—are now becoming progressively harder to cure. This compromises the quality of drug therapy, raises the risk of disease transmission, severe illness, disability, and death, while also wasting resources and increasing treatment costs³. In the United States, bacterial AMR in bloodstream infections was associated with 60,813 deaths, of which 14,987 were directly attributable, with the highest mortality rates observed in 2019⁴. Hosting over two-thirds of global inhabitants along with many low-income nations, the Asia-Pacific area has become a key focal point for AMR evolution and transmission⁵. Alarming, India has become one of the world's largest consumers of antibiotics, with escalating use across human health and agriculture, while simultaneously harboring some of the highest levels of multidrug-resistant pathogens, including tuberculosis⁶.

The global response has been substantial. The World Health Organization (WHO) initiated the Global Action Plan on AMR (2015), later endorsed by the United Nations, underscoring the urgency of coordinated strategies. The WHO's One Health framework recognizes the interconnectedness of human, animal, plant, and environmental health, promoting an integrated approach to containment⁷. WHO encourages member states to integrate complementary medicine into their national health policies as part of this strategy⁸. The Centers for Disease Control and Prevention (CDC) 2022 special report on COVID-

19's effect on AMR revealed that the pandemic strained healthcare facilities, health departments, and communities to their limits in 2020, severely hindering progress in addressing AMR⁹.

Recent data reveals that drug-resistant infections were directly linked to 1.27 million fatalities worldwide in 2019¹⁰. Antimicrobial resistance (AMR) is expected to lead to two million fatalities in India by 2050. India shoulders one of the most significant reservoirs of drug-resistant pathogens globally, encompassing the highest prevalence of multidrug-resistant tuberculosis¹¹. In India, the mortality rate associated with AMR surpasses that of fatalities resulting from respiratory infections, neoplasms, enteric diseases, tuberculosis, renal disorders, diabetes and maternal and neonatal conditions. The Tracking AMR Country Self-Assessment Survey (TrACSS) reports that India officially approved and implemented a National AMR action plan in 2017. India adopted its National Action Plan on AMR (NAP-AMR) 2017, outlining six strategic areas ranging from awareness-building and surveillance to optimizing antimicrobial use and fostering innovation. The following, six key strategic areas have been delineated within the NAP-AMR framework¹²:

Improving awareness and understanding of AMR through effective communication, education, and training.

Strengthening knowledge and evidence through surveillance. Reducing the incidence of infection through effective infection prevention and control.

Optimizing the use of antimicrobial agents in health, animals, and food. Promoting investments for AMR activities, research, and innovations; and strengthening India's leadership on AMR.

A critical factor in addressing AMR in a sustainable and enduring manner is 'Awareness.' Public initiatives such as the "Red Line Campaign", which marked antibiotic packaging with a red stripe to discourage misuse, further highlight efforts to raise awareness¹³. Despite these measures, AMR continues to escalate, with surveillance systems like the Global Antimicrobial Resistance and Use Surveillance System (GLASS) reporting alarming trends in antibiotic resistance globally. To curb the distribution of these medications, the Government of India has implemented stringent regulatory measures, including the introduction of Schedule H and H1 classifications¹².

Against this backdrop, there is growing recognition of the need for innovative alternatives to conventional antimicrobials—strategies that can prevent infections, slow the development of resistance, or enhance the effectiveness of existing drugs. Among these, homoeopathy has gained attention as a safe, cost-effective, and sustainable approach. Increasing evidence suggests that homoeopathic medicines may serve as supportive or even first-line interventions in the management of community-acquired infections, which are among the leading drivers of antibiotic consumption. Studies have reported that homoeopathy can stimulate immune responses, reduce recurrence rates, and, in some instances, perform comparably to standard therapies. Importantly, recurrence after homoeopathic intervention appears lower, indirectly curbing the need for repeated antibiotic use¹⁴.

This review aims to explore homoeopathy as a potential alternative and supportive therapy for community-acquired infections, thereby reducing dependence on antibiotics. Objectives include:

Describing the mechanisms of resistance and their global impact.

Examining pre-clinical and clinical evidence for homoeopathy in infections commonly treated with antibiotics.

Evaluating its role in reducing recurrence rates and antibiotic consumption.

Assessing its potential application in human, veterinary, and agricultural contexts within the One Health framework.

The novelty of this review lies in its emphasis on homoeopathy as an effective alternative strategy to reduce antibiotic dependence and mitigate AMR an area often overlooked in conventional discourse. Unlike most literature that focuses on stewardship, infection control, or drug discovery, this review highlights evidence from human, veterinary, and agricultural contexts to demonstrate homoeopathy's potential in lowering antibiotic use, reducing recurrence of infections, and minimizing environmental contamination.

A systematic literature search across PubMed, Google Scholar, BASE, and Science.gov identified pre-clinical and clinical studies on homoeopathy in respiratory, urinary, and gastrointestinal infections

commonly treated with antibiotics. Synthesizing this evidence demonstrates homoeopathy's potential to reduce recurrence, enhance immune response, and lower antibiotic reliance, offering insights relevant to both policymakers and clinicians in addressing AMR.

Considering the staggering global morbidity and mortality linked to AMR, particularly in India and the Asia-Pacific, integrating evidence-based homoeopathy into health frameworks is both timely and relevant. In line with the WHO's "One Health" approach, homoeopathy presents a promising yet underutilized strategy, though its role requires confirmation through large-scale, rigorous clinical trials to support policy adoption.

Materials and Methods

Literature searches were conducted on PubMed, Google Scholar, BASE and Science.gov, using the search terms "Antimicrobial resistance, Homoeopathy, Multi-drug-resistant infections, Rational drug use and Global health threats" to gather articles studying role of homoeopathy in antimicrobial resistance. Importance was given to selection of pre-clinical and clinical research studies on the effect of homoeopathic medicines in clinical conditions, veterinary, agricultural streams and while managing epidemic conditions. The search results were transferred to MS Excel, where duplicate entries were eliminated. Subsequently, titles and abstracts were meticulously screened, followed by a comprehensive evaluation of full-text articles to determine their eligibility.

Inclusion/Exclusion criteria

Studies included pre-clinical and clinical work, randomized trials, observational research, and case reports on homoeopathy in microbial diseases and AMR. Eligible publications were restricted to English-language articles between 1994 and 2024, covering respiratory, urinary, gastrointestinal, and systemic infections, as well as veterinary and agricultural contexts. Excluded were duplicates, non-retrievable texts, studies unrelated to AMR, those focused solely on antibiotics, abstracts without primary data, and articles outside the defined timeframe.

Results and Discussion

Prevalence of AMR

The discovery of antibiotics began in 1910 with salvarsan, the first of its kind. In 1928, Alexander Fleming identified penicillin, marking a major breakthrough. By the 1930s, Prontosil became the first antibacterial drug to be produced for commercial use. The era spanning the 1940s to the 1960s is frequently labeled as the "golden age" of antibiotics, marking the discovery of the majority of antibiotic categories that remain in use currently¹⁵. The discovery of penicillin was swiftly followed by the appearance of resistant organisms. By 1942, instances of *Staphylococcus aureus* resistant to penicillin had already been documented. The extensive use of antibiotics in agriculture during the 1950s and 1960s further hastened the development of resistance. The development of new antibiotics has significantly slowed since the mid-20th century. The WHO describes the current state of antibiotic research as bleak, with few promising breakthroughs on the horizon¹⁶.

Over the following decades, numerous types of antibiotics were introduced to combat bacteria through diverse mechanisms, each targeting specific bacterial structures or functions. They are classified based on their site of action and the way they disrupt bacterial processes¹⁷. Antibiotics exert their effects through six primary mechanisms: (1) disrupting cell wall formation, (2) blocking protein synthesis, (3) interfering with nucleic acid replication, (4) hindering essential metabolic pathways, (5) compromising membrane integrity, and (6) inhibiting ATP synthase activity (Fig. 1). Remarkably, it took about 20 years for resistance to develop against salvarsan¹⁸. However, resistance to sulfonamides and penicillin appeared much more rapidly, within roughly 12 years, as highlighted in the timeline below. Notably, biochemists Edward Abraham and Ernst Chain made a groundbreaking observation; they identified an enzyme from penicillin-resistant *Escherichia coli* capable of degrading penicillin, thereby neutralizing

its bacteria-killing properties¹⁹. Environmental resistance and its diversity reflect ancient antibiotic origins, highlighting the long-standing natural presence of such resistance. A targeted metagenomic analysis of permafrost samples from Beringia, aged 30,000 years, identified genes resistant to β -lactams, tetracyclines, and glycopeptides²⁰. Additionally, a comparison of genomes from modern isolates and permafrost strains dating back as far as 2.7 million years showed no notable variation in antibiotic resistance patterns between ancient and present-day pathogens²¹.

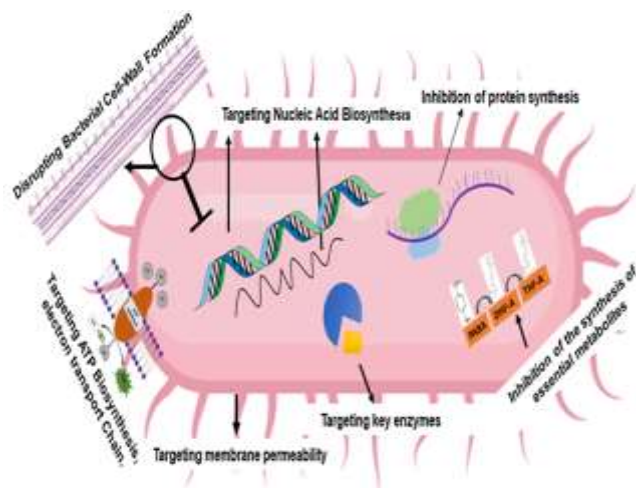


Figure 1: Modes of antibiotic action against bacteria

The widespread application of antibiotics in agriculture during the 1950s and 1960s played a pivotal role in accelerating AMR. The emergence of Methicillin-resistant *Staphylococcus aureus* (MRSA) in 1961 marked a significant milestone, soon followed by the development of resistance across multiple antibiotic classes²². In the 1980s, cases of multidrug-resistant (MDR) tuberculosis surged worldwide²³. Beginning in the mid-1960s, antimicrobial resistance, especially multidrug-resistance involving four or more antibiotics, became increasingly common in *S. typhimurium*. This trend accelerated significantly during the 1990s. The first documented instance of multi-drug resistance in *S. typhimurium* was reported in the U.K. in 1964. A strain (DT 29) was identified, displaying resistance to ampicillin, streptomycin, sulfonamides, tetracyclines, and furazolidone (R-type ASSuTFu) within the next five years. This resistant strain was responsible for widespread infections in humans, cattle, and particularly affected calves²⁴.

The rise of antibiotic-resistant gram-negative bacteria, particularly MDR strains, has raised significant concern, especially with the emergence of carbapenem resistance. Carbapenems were once regarded as the preferred therapy for severe infections caused by multidrug-resistant strains that produce extended-spectrum β -lactamases (ESBLs). However, the emergence of New Delhi metallo- β -lactamase-1 (NDM-1) has significantly compromised the effectiveness of carbapenems, creating a major healthcare threat¹³.

Jacoby et al.²⁵ reported the emergence of resistance to extended-spectrum beta-lactams (ESBLs) in gram-negative bacteria, including *Escherichia coli* and *Klebsiella pneumoniae*, by the 1990s. Resistance in *Escherichia coli* to ciprofloxacin, a frequently utilized antibiotic for urinary tract infections, was observed with prevalence rates spanning from 8.4% to 92.9%. Similarly, *Klebsiella pneumoniae* demonstrated resistance rates ranging from 4.1% to 79.4%. Penicillin resistance was reported in up to 51% of countries involved in the GLASS initiative²⁶. The increasing resistance rendered existing antibiotics less effective, prompting many pharmaceutical companies to discontinue their efforts in antibiotic research.

The WHO has released a revised list of priority pathogens, highlighting *Acinetobacter baumannii* as the most critical bacterium due to its remarkable ability to resist multiple antibiotics²⁷. Carbapenem-resistant *Acinetobacter baumannii* (CRAB) is particularly concerning due to its resistance to nearly all existing antibiotics, presenting a formidable

challenge in the context of environmental decontamination. CRAB is often linked to healthcare-associated infections, particularly in individuals using invasive medical devices or undergoing extended hospitalizations. Its ability to resist treatment stems from mechanisms like OXA-type carbapenemases, making many conventional antibiotics ineffective²⁸. A WHO report highlights that the pipeline for antibacterial treatments grew from 80 candidates in 2021 to 97 in 2023. However, just 12 of these are deemed novel, underscoring the ongoing scarcity of transformative therapies against the most critical bacterial threats²⁹. The current prevalence of AMR highlights a serious public health crisis, making the exploration of alternative therapies a crucial priority³⁰.

Causes and mechanism of resistance

AMR mainly arises from changes within microbes and can manifest through various mechanisms:

Genetic mutation

During bacterial replication, base pair changes may cause amino acid substitutions in key antibiotic targets like the cell wall, enzymes, or structures. These mutations can lower drug affinity, reduce uptake, or trigger efflux mechanisms to expel the drug³¹.

Transfer of genetic material: A once-susceptible strain can gain resistance by acquiring it from other species or genera. Resistance genes are commonly located on plasmids and mobile genetic elements, enabling their spread across various genera and species. Drug-resistant bacteria can pass copies of their genes to non-resistant bacteria, allowing the latter to acquire new DNA and develop resistance to drugs³².

Antimicrobial Selective Pressure: Selective pressure in antibiotic resistance occurs when antibiotics eliminate susceptible bacteria, enabling resistant bacteria to survive and proliferate. This process significantly contributes to the emergence of resistant strains. Essentially, selective pressure acts as a force in natural selection, favoring resistant organisms over others, driven by the presence of antibiotics³³.

The precise correlation between antibiotic usage and resistance remains unclear, but there is broad agreement that overuse plays a significant role in the development of antibiotic resistance. Studies indicate that an estimated one-fifth to one-half of antibiotic prescriptions in community healthcare settings are considered unnecessary. Furthermore, a Canadian survey revealed physicians compensated on a per-prescription basis exhibit nearly a fivefold increase in antibiotic prescribing rates compared to their peers. The main factors contributing to this issue include unregulated antibiotic use in healthcare settings, aggressive pharmaceutical marketing, and patient demand³⁴. Humans are key contributors to the development and spread of AMR through a range of activities, such as:

Excessive and improper use of antibiotics: Inaccurate diagnoses can lead to AMR by raising the chances of inappropriate antibiotic use. Healthcare professionals might prescribe broad-spectrum antibiotics rather than more appropriate, targeted alternatives. Additionally, misdiagnoses can lead to improper treatment measures that further exacerbate AMR³⁵. Self-medication can lead to irrational antibiotic use, including insufficient dosing, shortened treatment durations, and switching medications without doctor consultation. In many regions, the easy availability of antibiotics without prescriptions increases the risk of improper use, potentially causing adverse drug reactions, masking underlying diseases, and promoting microbial resistance.

Poor sanitation and hygiene in public spaces and hospitals

Inadequate infection control practices in hospitals can facilitate the spread of AMR, as pathogenic germs can transfer between individuals via surfaces such as wheelchairs, bedrails, medical diagnostic equipment's or the hands of healthcare workers. Ultimately, these factors lead to the continued development and spread of microbial strains resistant to various antibiotics, posing a significant threat to public health. Additionally, discharge of hospital water and effluents without proper sterilization disseminate antimicrobial-resistant

organisms and genes into the surrounding environment mainly the water bodies, facilitating their emergence and increasing the likelihood of human and animal exposure living in the environment³⁶.

Developing new antibiotics is not the ultimate solution

The persistent issue of antibiotic resistance remains unresolved, even with the introduction of new antibiotics. Over the past decade, resistance among respiratory bacterial pathogens has surged, while advancements in developing effective antibiotic agents have been modest. Newly licensed antibiotics must be used sparingly due to their significant side effects, high costs, and potential to exacerbate resistance. The root problem lies in bacteria's ability to rapidly adapt, making resistance to new drugs inevitable and perpetuating a cycle of costly drug development.

The misuse and overuse of existing antibiotics remain the primary drivers of AMR, accelerating the rise of resistant bacteria and diminishing the efficacy of new treatments. Rapid diagnostic technologies that accurately identify infections requiring antibiotics could reduce unnecessary usage, but pharmaceutical companies face limited financial incentives to develop novel antibiotics due to high costs and the likelihood of rapid resistance development³⁷. In light of these challenges, alternative therapeutic approaches hold significant promise. Strategies focusing on disrupting bacterial virulence mechanisms or enhancing host immunity, rather than directly eradicating bacteria, could provide a more sustainable and effective solution to the escalating AMR crisis.

Combating antimicrobial resistance: Homoeopathy an evidence-based approach

Antibiotic use is the main driver of AMR, with over 50% considered unjustified. Irresponsible usage occurs in humans, veterinary practices, and animal husbandry for both therapeutic and non-therapeutic reasons, such as growth promotion³⁸. Antimicrobials are often misused instead of adhering to proper asepsis, cleanliness, and infection prevention and control (IPC) in communities, hospitals, and farms, contributing to the spread of MDR pathogens. The extent of antibiotic abuse is similar in humans and animals. Due to the rise in AMR and the shortage of new, effective antibiotics, there is increasing interest in conducting clinical research to validate traditional medicines.

Hahnemann criticized the conventional medicine of the late 18th century for being irrational and ineffective, often causing harm rather than providing benefits. He proposed an alternative approach, advocating for the use of single, lower-dose medications. Additionally, Hahnemann emphasized a vitalistic, non-materialistic view of life, suggesting that living organisms operate through an unseen, vital force. This philosophy underpinned his development of homoeopathy, designed to stimulate the body's innate healing abilities³⁹. Homoeopathy is a well-established form of complementary and alternative medicine (CAM) used globally, known for its excellent safety record. Homoeopathy has been an integral part of European healthcare for over two centuries. A report from the European Council for Classical Homoeopathy (ECCH) revealed that the practice is widely embraced by professionals throughout Europe, and homoeopathic remedies are readily accessible to the public as non-prescription products⁴⁰.

The preventive application of homoeopathy dates back to 1801, amidst a scarlet fever epidemic in Königslutter, Germany. In this case, Samuel Hahnemann, the founder of homoeopathy, prescribed a single dose of *Belladonna* to children deemed susceptible to the disease. This intervention proved highly successful, with a reported 100% effectiveness rate among those treated⁴¹. This early use of homoeopathy in epidemic prevention is often considered a foundational moment in the development of homoeopathic practices for infectious disease management. Research into the physico-chemical and nanoparticulate properties of homoeopathic remedies, as well as their adaptive nonlinear responses in living systems at low doses, supports its effectiveness as an alternative therapy⁴². Moreover, a steadily increasing volume of clinical research, featuring comparative effectiveness analyses involving thousands of homoeopathic patients, alongside data on safety and cost-effectiveness, highlights

homoeopathy as a credible therapeutic option. Homoeopathy medicines are known to influence immune function at multiple levels. This included altering gene expression related to receptor and protein production in macrophages, stimulating macrophages as indicated by morphological changes, modifying receptor expression on their surface, promoting polymorphonuclear (PMN) leukocytes cell chemotaxis, and inducing cytokine release along with reactive nitrogen and oxygen species production⁴³⁻⁴⁵ (Figure 2).

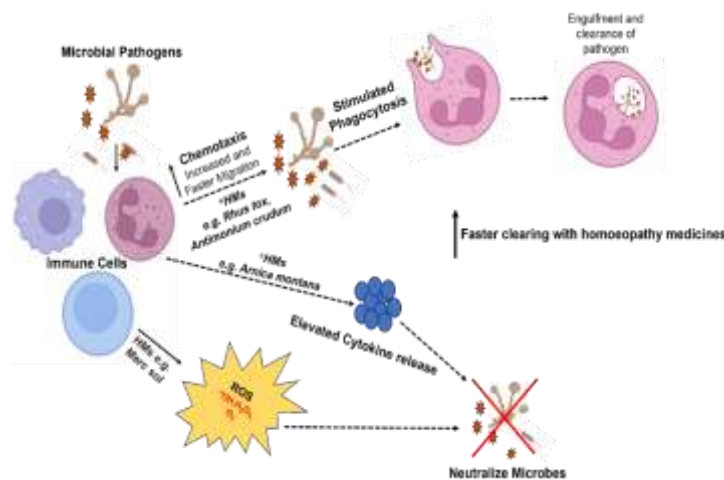


Figure 2: Immunomodulatory effects of homoeopathic medicines⁴³⁻⁴⁵

Approximately half of tonsillitis cases are linked to group-A beta-hemolytic *streptococci*, which are generally susceptible; however, many strains have already developed resistance to commonly used antibiotics, such as penicillin and macrolides. These antibiotics can be easily replaced by homoeopathic alternative *Apis mellifica* which has shown promising results since olden times⁴⁶. Our literature search on investigating the role of homoeopathy in AMR resulted in the studies including seven pre-clinical studies and ten clinical studies on Ear, Nose, Throat (ENT) or Upper Respiratory Tract Infections (URTIs), with three Urinary tract infections, and three Gastro-intestinal tract infections as comprised under Table 1 and 2. The studies reflected a positive outcome of homoeopathy as a standalone or add on to conventional treatment. The clinical research data suggested that homoeopathy is effective in the treatment of URTIs in children which is a frequent cause of inappropriate antibiotic prescribing. A review comprises of 29 studies in the domain 'URTIs/ allergic reactions' showed a positive overall result in favor of homoeopathy. Applied properly, classical homoeopathy has few side-effects, and the use of high potencies is free of toxic effects⁴⁷. A comprehensive analysis of clinical studies conducted after 1994 on homoeopathic treatments for acute URTIs and related complications highlights that, given the rise of antimicrobial resistance, homoeopathy provides alternative approaches for managing minor infections and potentially preventing recurrent URTIs. Leading epidemiologists propose that homoeopathy exhibits significant efficacy in managing respiratory tract infections, yielding outcomes comparable to conventional medical treatments in mitigating severe complications. A thorough analysis of moderate to high-quality evidence supports the beneficial role of homoeopathy in URTIs, especially in conditions frequently managed with antimicrobial medications, such as otitis media, sinusitis, the common cold, influenza-like syndromes, and unspecified URTIs.

Approximately three-quarters of all antibiotics are administered to treat respiratory tract infections. Although antibiotic use significantly reduces this risk, the number needed to treat to prevent one serious complication exceeds 4,000 across all these conditions leading to extensive use which further accelerates the possibility of resistance⁴⁸. If homoeopathy is used as the first line of treatment in treating URTIs, especially in its initial stages, we may cut down 75% of the use of antibiotics, thereby combating AMR. Some of the studies also suggested that the recurrence of infection is very less after

homoeopathic intervention, which also reduces the use of antibiotics. *In-vitro* studies also showed significant results using homoeopathic medicines as compared to antibiotics in modulating the immune mechanisms of patients and bringing back general well-being⁴⁹. A list

of a few pre-clinical and clinical studies showing role of homoeopathy on drug-resistant pathogens or diseases caused by them is listed in Table No. 1 and 2.

Table 1: List of a few pre-clinical studies on the role of homoeopathy on drug-resistant pathogens

S/N	Type of infections/ Pathogens	Studies conducted	Outcome
1.	<i>Staphylococcus aureus</i>	<i>Staphylococcus aureus</i> isolated from clinical samples was analyzed in a comparative study to evaluate the efficacy of homoeopathic and allopathic treatments.	This <i>in-vitro</i> study revealed that <i>Belladonna</i> 200C, <i>Arnica</i> 200C, <i>Nux Vomica</i> 200C, <i>Sulphur</i> 200C, and <i>Berberis Vulgaris</i> 200C possess notable antimicrobial activity against resistant bacterial strains like <i>Staphylococcus aureus</i> . Moreover, these homoeopathic preparations offer a potential alternative to conventional antibiotics, as they neither weaken the immune system nor promote antibiotic resistance ⁷² .
2.	<i>Staphylococcus aureus</i> .	Antimicrobial efficacy of homoeopathic preparations against <i>Staphylococcus aureus</i> : An <i>in-vitro</i> study	In an <i>in vitro</i> study demonstrated that <i>Lachesis</i> and <i>Staphylococcinum</i> nosode significantly ($p < 0.05$) inhibited the growth of <i>Staphylococcus aureus</i> ⁷³ .
3.	<i>Staphylococcus aureus</i> .	Impact of <i>Oxacillin</i> on the <i>in-vitro</i> proliferation of methicillin-resistant <i>Staphylococcus aureus</i> (MRSA) following treatment with homoeopathic remedies	This study demonstrated that methicillin-resistant <i>Staphylococcus aureus</i> (MRSA) cultures exposed to <i>Belladonna</i> or <i>MRSA</i> nosode exhibited a reduction in growth, a decrease in enzymatic activity, and an increased susceptibility to the antibiotic oxacillin. These findings underscore the potential influence of homoeopathic treatments on antibiotic-resistant bacteria ⁷⁴ .
4.	<i>Enterococcus faecalis</i>	Homoeopathic consideration for resistant endodontic bacteria <i>Enterococcus faecalis</i> : An <i>in vitro</i> comparative disc diffusion study	A study investigating endodontic infections caused by <i>Enterococcus faecalis</i> , which is responsible for around 40% of primary cases and up to 77% of persistent or secondary infections, demonstrated that the homoeopathic remedy <i>Benzoicum Acid</i> exhibited the strongest inhibitory effect against <i>E. faecalis</i> . This was followed by <i>Silicea</i> extract, with calcium hydroxide showing comparatively lower effectiveness in expanding the zone of inhibition against this Gram-positive bacterium ⁷⁵ .
5.	<i>Escherichia coli</i> , <i>Salmonella sp.</i> and <i>Vibrio sp.</i>	Evaluation of antimicrobial activity and determination of suitable method for antibiogram of some commercial homoeopathic drugs	The study identified the inhibitory concentration range of certain homoeopathic medicines, demonstrating their antibacterial potential against clinically isolated microorganisms, including <i>Escherichia coli</i> , <i>Salmonella sp.</i> , and <i>Vibrio sp.</i> ⁷⁶ .
6.	<i>Escherichia coli</i>	An <i>in-vitro</i> study of antimicrobial activity of homoeopathic medicine <i>Pyrogenium</i> against <i>Escherichia coli</i> .	<i>Pyrogenium</i> 30C produced a modest to moderate inhibition zone, restricting <i>Escherichia coli</i> growth by 7.14% ⁷⁷ .
7.	<i>Escherichia coli</i> , <i>Klebsiella pneumoniae</i> , <i>Salmonella typhi</i> , <i>Neisseria gonorrhoeae</i> , and <i>Candida albicans</i>	<i>In-vitro</i> evaluation of antimicrobial activities of <i>Escherichia coli</i> , <i>Klebsiella pneumoniae</i> , <i>Salmonella typhi</i> , <i>Neisseria</i>	The investigation revealed that the tested nosodes exhibited notable antimicrobial activity. Specifically, <i>C. albicans</i> polyvalent nosode 35C and 100C, along with <i>N. gonorrhoeae</i> 35C and the reference antifungal amphotericin B, effectively inhibited the proliferation of <i>C. albicans</i> species. Similarly, <i>K. pneumoniae</i> 35C and <i>E. coli</i> polyvalent nosode 100C, alongside the standard antibiotic meropenem,

- gonorrhoeae*, and *Candida albicans* Nosodes. demonstrated inhibitory effects on *K. pneumoniae*. Furthermore, *E. coli* polyvalent nosode 30C in 10% alcohol (both undiluted and at a 1:1 dilution) suppressed the growth of *E. coli*, akin to the positive control's ciprofloxacin, ofloxacin, and amoxicillin. Additionally, the *S. typhi* polyvalent nosode 30C in 10% alcohol exhibited antibacterial activity not only against *S. typhi* but also against other selected pathogens, highlighting its broader antimicrobial efficacy⁷⁸.
8. Uropathogenic *E. coli* strain JJ079 To investigate the impact of *Cantharis* 6CH, a homoeopathic remedy, on cystitis caused by *E. coli* infection. The findings of this study demonstrated a 50% decrease in the incidence of erythematous cystitis in animals treated with *Cantharis* 6CH, as independently confirmed by two observers during necropsy. This improvement was accompanied by notable shifts in immune cell distribution across various regions of the urinary tract⁷⁹.
9. *Encephalitozoon cuniculi* To investigate whether macrophages infected with *E. cuniculi* exhibit functional changes following treatment with varying potencies of *Phosphorus* in an *in-vitro* model. Treatment with the highest potency of *Phosphorus* (Phos 200CH) uniquely triggered a marked, time-dependent escalation in RANTES (regulated on activation, normal T-cell expressed and secreted) levels and lysosomal activity ($p \leq 0.002$). This was coupled with a significant decline in apoptosis, intensified digestion of parasites, and the persistence of spores remaining outside host cells⁸⁰.
10. *Leishmania amazonensis* To investigate the impact of M1, a complex homoeopathic formulation, on the interaction between macrophages and *Leishmania* both *in-vitro* and *in-vivo*. The findings of this study demonstrate that the highly diluted M1 formulation of homoeopathic medicines elicits rapid and effective biological responses against *Leishmania* in living organisms, cells, and mice. These effects are evident through alterations in cytokine profiles, a significant increase in nitric oxide production, a marked reduction in parasitic load, and modifications in the classical maturation and biogenesis of parasitophorous vacuoles. Additionally, the M1 complex significantly reduced the endocytic index ($p < 0.001$) and the percentage of infected macrophages ($p < 0.05$), effectively preventing lesion development caused by *L. amazonensis* by enhancing the Th1 immune response⁸¹.
11. *Trypanosoma cruzi* To investigate the impact of *Lycopodium clavatum*, prepared at a 13C dynamization, on Wistar rats experimentally infected with *T. cruzi*. the findings demonstrated that *L. clavatum* 13C exerted a positive immunomodulatory effect in rats infected with *T. cruzi*, effectively mitigating the pathological progression of digestive Chagas disease⁸².
12. *Leishmania infantum* To evaluate the effects of treatment with homoeopathy medicine BioLi30X followed by post-treatment with *Antimonium crudum* 30X on the dominance of Th1/Th2 immune responses, spleen morphology changes, and megakaryocyte dynamics in BALB/c mice infected with *Leishmania infantum*. Pre/post-treatment with *BioLi30* and post-treatment with *Antimonium crudum* 30 demonstrated greater effectiveness in controlling the disease. These findings suggest their potential to modulate the immune system, steering it toward a more favorable prognosis characterized by a Th1 response⁸³.

Table 2: List of a few clinical studies on the role of homoeopathy on drug-resistant pathogens

S.No.	Type of infections/ Pathogens	Studies conducted	Outcome
1.	ENT and Upper Respiratory Tract Infections: (Otitis media, sinusitis, common cold, influenza-like syndromes, and (unspecified) upper respiratory tract infections) <i>Streptococcus pneumonia</i> <i>Haemophilus influenzae</i> <i>Pseudomonas aeruginosa</i> <i>Mycobacterium</i> <i>tuberculosis</i>	1. Effects of homoeopathic medicines on daily burden of symptoms in children with recurrent upper respiratory tract infections. 2. Homoeopathic medicine for acute cough in upper respiratory tract infections and acute bronchitis: a randomized, double-blind, placebo-controlled trial. 3. Homoeopathy in acute rhinosinusitis: a double-blind, placebo-controlled study shows the efficiency and tolerability of a homoeopathic combination remedy. 4. A randomized placebo controlled clinical trial using homoeopathy as an adjuvant to standard care in the viral infections 5. <i>Oscillocoffin®</i> in patients with influenza-like syndromes: a placebo-controlled double-blind evaluation. 6. Effect of individualized homoeopathic treatment in influenza like illness: A multicenter, single blind, randomized, placebo-controlled study. 7. Randomized controlled pilot study to compare homoeopathy and	In a controlled clinical trial with a double-blind and placebo-based design, antibiotic usage saw a notable decline compared to the previous year, decreasing from 73 to 33 in the homoeopathic treatment group and from 69 to 43 in the placebo group. Additionally, a lower percentage of children in the treatment group underwent adenoidectomy (16%, 8 out of 50) compared to those in the placebo group (21%, 9 out of 42). However, the tonsillectomy rate remained unchanged at 5% in both groups⁸⁴. After 4 and 7 days of treatment, this double-blind, randomized, placebo-controlled trial revealed a significant reduction in cough severity in the homoeopathic group compared to the placebo group. The study concluded that the homoeopathic syrup administered effectively decreased both cough severity and sputum viscosity, making it a promising option for managing acute cough associated with URTIs⁸⁵. A clinical trial employing a randomized, double-blind design investigated the effects of homoeopathic treatment in individuals with acute rhinosinusitis. The findings revealed that 90.3% of patients receiving homoeopathy experienced complete resolution of symptoms, while an additional 8.3% showed improvement. In contrast, 88.9% of those in the placebo group exhibited no change or a worsening of their condition⁸⁶. In a randomized placebo-controlled trial on COVID-19, individualized remedies including <i>Bryonia alba</i>, <i>Arsenicum album</i>, <i>Pulsatilla</i>, and <i>Camphora</i> led to faster relief in cough, breathlessness, weakness, and fatigue, while significantly reducing ICU admission, ventilator use, and mortality compared to standard care alone⁸⁷. This clinical trial demonstrated that after 48 hours, patients in the verum group experienced significantly milder symptoms (P=0.023) compared to those in the placebo group. Additionally, from the second day onward, the proportion of asymptomatic patients was notably higher in the verum group (17.4%) than in the placebo group (6.6%). These findings indicate that treatment with <i>Oscillocoffin®</i> has a beneficial impact on reducing symptom severity and shortening the duration of influenza-like syndromes⁸⁸. A randomized, triple-arm, placebo-controlled multicenter study demonstrated the significant efficacy of individualized homoeopathic treatment in managing influenza-like illness. The outcomes showed no notable difference between the LM and Centesimal potency groups. Furthermore, the intervention groups exhibited a significantly lower rate of complications and sequelae⁸⁹. By the third day of treatment, a pilot study with a randomized, placebo-controlled parallel-group design found that four patients in the homoeopathy group had recovered, whereas only one patient recovered in the conventional treatment group.

conventional therapy in acute otitis media (AOM). Furthermore, antibiotics were prescribed to 39 patients (97.5%) in the conventional group, while none were required in the homoeopathy group⁹⁰.

8. Homoeopathic ear drops as an adjunct to standard therapy in children with AOM. Homoeopathic ear drops demonstrated moderate effectiveness in alleviating otalgia among children with AOM. Their benefits appeared to be most pronounced in the early stages following an AOM diagnosis⁹¹.

9. Homoeopathic treatment of acute otitis media in children: a preliminary randomized placebo-controlled trial. Children receiving individualized homoeopathic medicine experienced fewer treatment failures after 5 days, 2 weeks, and 6 weeks, with observed differences of 11.4%, 18.4%, and 19.9%, respectively, though these differences were not statistically significant. Additionally, diary scores revealed a significant reduction in symptoms at 24 and 64 hours post-treatment in favor of homoeopathy ($P < 0.05$). These findings indicate that there is a positive effect of homoeopathy in acute otitis media, compared to placebo⁹².

10. Homoeopathic treatment in addition to standard care in multidrug-resistant pulmonary tuberculosis: a randomized, double blind, placebo controlled clinical trial. Among 120 patients diagnosed with MDR-TB, those treated with a standard regimen combined with individualized homoeopathic medicine achieved an 11.4% higher cure rate compared to those in the placebo group, as observed in a double-blind, placebo-controlled study⁹³.

2. Urinary Tract Infections (UTIs)
Escherichia coli,
Klebsiella pneumoniae,
Proteus mirabilis,
Enterococcus faecalis
Staphylococcus saprophyticus
 1. A dual case examination of integrated strategies for antibiotic-resistant UTIs
This study shows two case reports presenting that homoeopathy not only has cured but also limited antibiotic course, improved their quality of life and prevented relapses too⁹⁴.
 2. Efficacy of *Chimaphila umbellata* in the treatment of UTIs in adolescent girls
A study utilizing a single-blind randomized design demonstrated that *Chimaphila umbellata* exhibits significant effectiveness in managing UTIs among adolescent girls⁹⁵.
 - A single-blind randomized control study.
This report presents four female cases of recurrent UTIs that were successfully managed using a homoeopathic approach. The indicated remedies—*Lac caninum*, *Pulsatilla*, *Causticum*, and *Arsenicum album*—proved effective after conventional treatments failed to yield improvement⁹⁶.
3. Gastro-intestinal infections: Diarrhoea
Escherichia coli
Colibacillosis
 1. Homoeopathic approaches to calf diarrhea: *Echinacea angustifolia*'s clinical edge
A study investigating the use of *Echinacea angustifolia* to address naturally occurring neonatal calf diarrhea, while mitigating the adverse effects of indiscriminate antibiotic use in dairy herds, revealed that clinical recovery was faster in the group treated with *Echinacea angustifolia* compared to other groups, including those receiving standard antibiotic treatments⁹⁷.
 2. Evaluating homoeopathy as an alternative to antibiotics
This study explored the use of homoeopathy as an alternative to antibiotics for managing *Escherichia coli* diarrhea in neonatal piglets. Results showed that piglets

for *Escherichia coli* treated with homoeopathy experienced significantly fewer cases of *Escherichia coli* diarrhoea in neonatal piglets compared to those in the placebo group ($P < 0.0001$)⁹⁸.

3. Assessing the effectiveness of preventive homoeopathic interventions for colibacillosis in swine production. Treatments using homoeopathy and bio-therapies proved more effective than antibiotics in managing diarrhea among newborn piglets. The combined approach of homoeopathic and isopathic therapies resulted in the greatest weight gain. These findings highlight homoeopathy and isopathy as viable alternatives for addressing *Escherichia coli* induced diarrhea in newborn swine⁹⁹.

Historically, homoeopathy was a trusted method for managing epidemics before the advent of vaccines. While vaccines revolutionized disease prevention and effectively addressed emerging resistant microbial strains, their efficacy has gradually diminished over time, prompting a re-evaluation of alternative approaches like homoeopathy. At this stage, homoeopathy emerges as a viable and explorable field of epidemics with distinct benefits such as its easy method of preparation, storage, and higher level of safety, etc. So many studies have been conducted on Epidemic diseases where homoeopathy acted as prophylactic. During a leptospirosis epidemic in Cuba, a homoeopathic prophylactic formulation was orally administered to 2.3 million high-risk individuals and demonstrated effectiveness⁵⁰. In India, research on Chikungunya fever and Dengue found that homoeopathic prophylactic medicines were effective, highlighting their potential as a practical approach to managing epidemic outbreaks^{51, 52}.

Homoeopathy is gaining recognition as a legitimate alternative therapy in veterinary medicine, valued for its capacity to prevent disease recurrence without leaving residual substances in animal-derived products. A specific study has shown that homoeopathic metaphylaxis is considerably more effective than both placebo and conventional low-dose antibiotic metaphylaxis in lowering the incidence and progression of respiratory tract diseases in the evaluated animals⁵³. Weiermayer et al.⁵⁴ provided a definitive report presenting an overview of homoeopathy and veterinary homoeopathy, highlighting its potential role in addressing the AMR problem.

The application of antibiotics in agriculture leads to environmental contamination across various reservoirs, creating conditions that favor the emergence and propagation of resistant microbes and microbial genes⁵⁵.

More than half of the global antibiotic production is reportedly utilized by the farming industry²⁷. Homoeopathy provides a sustainable alternative to antibiotics and is becoming increasingly popular in the agricultural sector. Endorsed by the European Union for use in organic animal husbandry, it has been widely adopted by many farmers. The adoption of homoeopathy has led to an approximate 50% reduction in medication expenses⁴⁷. A range of synthetic fungicides is utilized to control *Aspergillus niger* contamination in food and various environments. However, *Aspergillus niger* has evolved resistance to antifungal agents and is regarded as a “model organism” for assessing the antifungal efficacy of different compounds⁵⁶. Prajapati et al.⁵⁷ have demonstrated efficient antifungal activity of Homoeopathic medicines by inhibiting growth of *Aspergillus niger*.

This review underscores the pressing need for alternative strategies such as homoeopathy in combating AMR, a rapidly escalating global health threat. Once resistance develops to specific microorganisms or drugs, current treatment options become severely limited, often requiring repeated antibiotic courses with little improvement. AMR is now recognized as a major public health crisis worldwide, with a rapidly increasing burden documented through statistical evidence. In the U.K. alone, 178 AMR infections were diagnosed daily in 2019, an increase of eight cases per day compared to 2018⁵⁸. Similarly, the U.S. CDC reported a 20% surge in six antimicrobial-resistant, hospital-onset bacterial infections during the COVID-19 pandemic, peaking in 2021 and persisting above pre-pandemic levels in 2022⁵⁹. Japan recorded around 8,000 AMR-related deaths in 2017 from MRSA and fluoroquinolone-resistant *Escherichia coli* (FQREC). Despite

implementing a containment plan aiming to reduce MRSA resistance to 20% by 2020, the country achieved only a modest annual reduction of 2%⁶⁰. Australia, too, reported 1,600 direct AMR related deaths and 6,700 associated deaths in 2019⁶¹. The mortality burden was even more pronounced in India, with 297,000 deaths directly attributed to AMR and over 1,042,500 associated deaths in 2019¹⁰. China also experienced 145,500 deaths linked to AMR and 602,600 deaths associated with AMR in 2019⁶². Collectively, these figures highlight the staggering toll of AMR across continents and underscore the urgent need for alternative interventions.

In this context, homoeopathy emerges as a potentially valuable tool. It is widely practiced across the globe, with its popularity growing in response to side effects and resistance associated with conventional medicines. The U.K.'s Five-Year AMR Action Plan encourages exploring multi-faceted approaches, with the Homoeopathy Research Institute initiating clinical trials in this field⁶³.

Homoeopathy, an alternative medicine system, has experienced renewed public interest in recent years. In the U.S., homoeopathic medicines are typically regulated by the Food and Drug Administration (FDA) and are frequently used as complementary and integrative therapies⁶⁴. Homoeopathy is the most widely used alternative therapy in five European countries and ranks among the top three in 11, according to a Norwegian survey⁶⁵. Surveys indicate that 20–25% of Europeans use homoeopathy, with coverage included in some insurance schemes⁶⁶. Usage rates have risen notably; in France, the prevalence increased from 16% in 1982 to 36% in 1992⁶⁷. Homoeopathy is also the leading complementary therapy in Belgium, the Netherlands, Norway, and Switzerland⁶⁶. An Italian study found 8.2% of the population, including 7.7% of children, use homoeopathy, primarily due to concerns over conventional medicine's toxicity⁶⁸.

Recently discovered antibiotics have not been effective in curbing the rising resistance in bacterial pathogens, with limited progress in developing new treatments. Caution is essential when using newly approved antibiotics due to side effects, high costs, and the risk of worsening resistance. Given the potential for serious complications, urgent preventive action and collaboration among governments, pharmaceutical companies, healthcare providers, and the public are necessary. In response, the European Union (EU) is developing strategies to promote responsible antibiotic use, focusing on surveillance, education, prevention, and collaborative research with other government officials⁶⁹. As homoeopathy doesn't depend on any kind of specific strain and it works on symptom similarity, the individual as a whole is taken into consideration, thereby strengthening the immune system. This could lead to shorter hospital stays, reduced healthcare expenses, fewer missed days of work, and better adherence to conventional treatments³⁹. Combining homoeopathy with conventional medical care might potentially aid in the treatment of various infectious diseases, particularly those with high mortality and morbidity rates. Homoeopathy, although typically cost-effective, may see its affordability diminished due to the necessity for longer or more frequent consultations. Furthermore, the costs can differ between countries and depend on the specific reimbursement systems in place. Homoeopathic medicines can be efficiently mass-produced, and their straightforward distribution makes them a practical choice for use during epidemics and pandemics.

Clinical outcomes provide encouraging comparative insights. Adler et al.⁷⁰ reported a 4.4% reduction in hospitalization rates during COVID-19 among patients treated with the homeopathic remedy *Natrum muriaticum* LM2. Similarly, Mahesh et al.⁷¹ documented a significant recovery rate of 73% in COVID-19 patients who received classical homeopathy treatment. Such evidence suggests that, beyond its role as supportive care, homeopathy may strengthen immunity, reduce infectious episodes, shorten hospital stays, and lower overall healthcare costs. Unlike antibiotics, it works on the principle of symptom similarity, treating the individual holistically rather than targeting specific microbial strains. Homeopathic medicines have consistently demonstrated a strong safety profile, with meta-analyses confirming the absence of serious adverse effects across studies. It is suggested that by enabling the body to fight infections naturally, homeopathy enhances immune resilience, a mechanism that contrasts with the dependency and resistance risks associated with antibiotics (Fig. 3)^{39, 70-71}. Furthermore, homeopathic remedies are free from serious side effects due to their non-toxic nature. A meta-analysis of homeopathic trials confirmed the absence of serious side effects across all included studies.



Figure 3: Comparison of the mode of action of homeopathic medicines and antibiotics^{39, 70-71}.

Conclusion

The rapid emergence of AMR and the shortcomings of conventional therapies underscore the need for alternative and complementary solutions. Homeopathy, with its individualized treatment approach and favorable safety profile, offers a promising avenue. Future research should prioritize rigorously designed pre-clinical and large-scale clinical trials to evaluate the effectiveness of homeopathic medicines against drug-resistant microbes, including multidrug-resistant strains of major infectious diseases. Generating high-quality, reproducible evidence could not only validate homeopathy's therapeutic role but also support its integration into mainstream healthcare. Importantly, such evidence may guide the incorporation of homeopathy into the WHO's "One Health" framework and national AMR action plans, particularly in regions heavily burdened by resistance. By complementing stewardship, infection control, and drug development strategies, homeopathy has the potential to strengthen global AMR containment policies and contribute to sustainable public health solutions.

Conflicting Interest

The authors declare no conflict of interest.

Authors Declarations

The authors hereby declare that the work presented in this article is original and that any liability for claims relating to the content of this article will be borne by them.

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